

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2022-23

Paper No. BSC-07A: Ancillary Physics-II

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **four** of the following: [4 × 3.5 = 14]

- (a) State Fermat's principle of least time with an example.
- (b) What is polarized light? What nature of light is shown by polarization?
- (c) What will be the refractive index of a medium in which the speed of light is 2.5×10^8 m/s?
- (d) Give an innovative example to explain entropy.
- (e) An ideal heat engine operating between the temperatures T_1 and T_2 ($T_1 > T_2$) has efficiency η . What would be the efficiency of the engine if you double the operating temperatures?
- (f) Deduce the unit of Boltzmann's constant from the ideal gas equation.

Question 2.

- (a) How do you define heat? Explain the heat capacity at constant volume and constant pressure. [6]
- (b) Write the formulae and use them to convert the boiling temperature of gold, 2966°C , into degrees Fahrenheit and Kelvin. [6]
- (c) What is the unit of pressure in the SI system of units? [2]

Question 3.

- (a) Define Viscosity and Thermal conductivity and give an example for each. [6]
- (b) The heat capacity of 0.125 kg of water is measured to be 523 J/K at room temperature. Calculate the heat capacity of water per unit mass and per unit volume. [5]
- (c) Which law of thermodynamics makes sense in the biological process of Photosynthesis? Explain. [3]

Question 4.

- (a) What is thermal equilibrium? Use it to explain the Zeroth law of thermodynamics. [6]
- (b) What is the relation between free energy and work? Starting from $F = U - TS$, where F is the Helmholtz function and other notations carry the usual meaning, express the entropy and pressure at constant volume and temperature, respectively. [6]
- (c) What are the state variables and how do you label different thermodynamic processes by fixing one of these variables? [2]
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Question 5.

- (a) Explain the differences between the following:
- (i) Concave and Convex mirrors
 - (ii) Interference and Diffraction
 - (iii) Real image and Virtual image
- [6]
- (b) Explain Young's double-slit experiment with a neat ray diagram. Write the expression for the fringe-width of interference fringes. Why are all fringes of the same width? [6]
- (c) An object is kept at a position between the optical center and the focus of a convex lens. Show the image formation with the help of a neat diagram and write the properties of the image formed. [2]
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Question 6.

- (a) Explain the optical phenomena involved in the following statements:
- (i) Meters like ammeters and voltmeters use a mirror to avoid parallax error.
 - (ii) A diamond sparkles more than a glass piece of the same shape and size.
 - (iii) A cat's eye or road stud helps drivers by improving night visibility.
- [6]
- (b) In the dispersion of light by a prism, which color is more dispersed? Explain quantitatively using the formula of minimum deviation angle. [5]
- (c) If the critical angle of a medium is 30° , find the polarizing angle for the medium. [3]
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Question 7.

- (a) In the figure shown below, when light falls on the vertical surface BC , it refracts along the direction BC . Find the incidence angle i . Given $\sin^{-1}(0.75) = 48.6^\circ$. [5]
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- (b) What is farsightedness or hyperopia? How can it be removed? Explain with the help of a neat diagram. [5]
- (c) Find the critical angle for a ray of light passing from glass ($\mu_g = 1.5$) to water ($\mu_w = 1.33$). [4]